MISR LANGUAGE SCHOOLS
National Primary Stage

Wonders

Leveled Readers



Name: _____

Class : _____

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MISR LANGUAGE SCHOOLS National Primary Stage

English Language Arts

Primary V

Wonders

Leveled Readers

School Year 2026/2027

Name:_____

Class : _____

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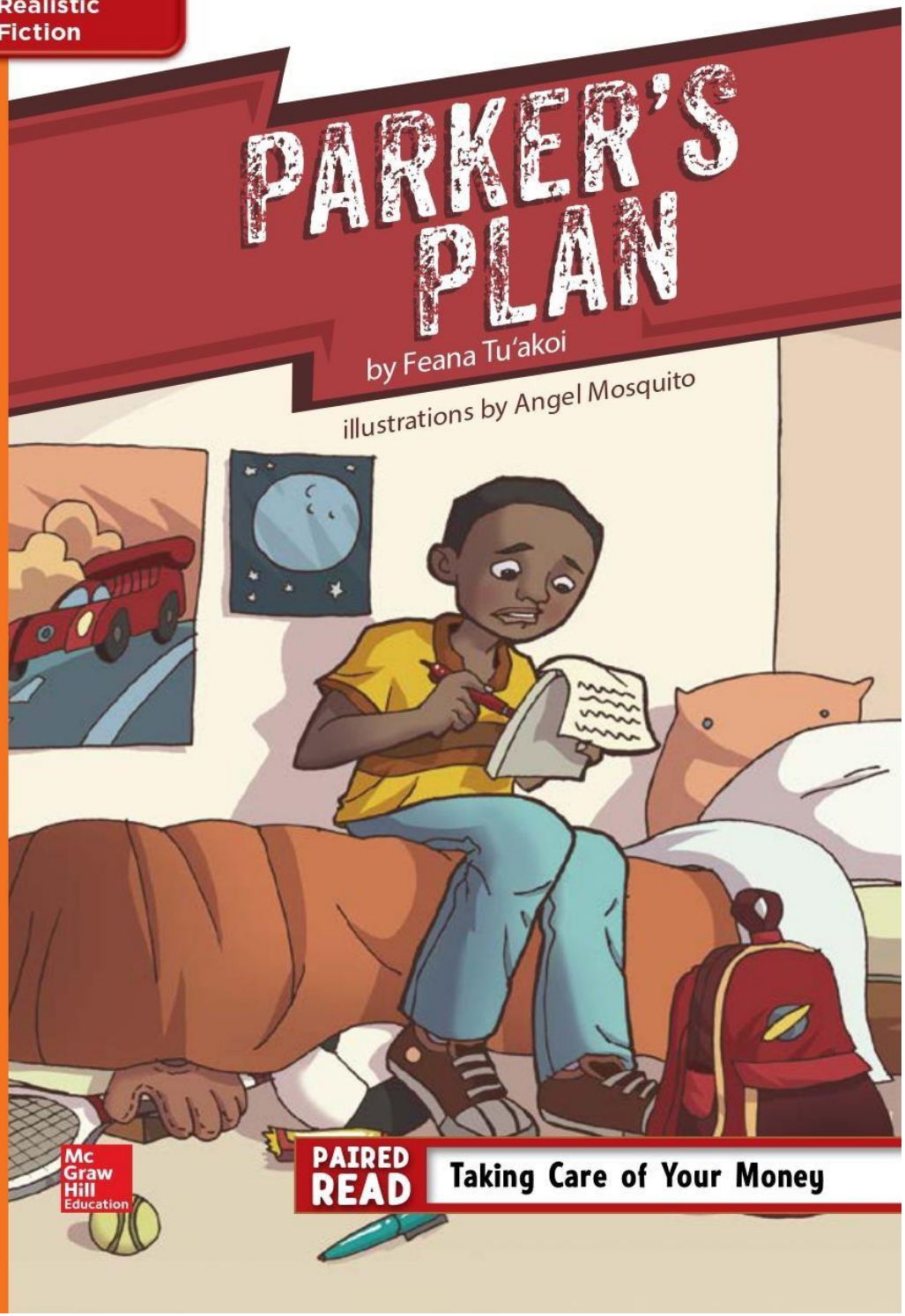
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Realistic Fiction



Genre Realistic Fiction

Essential Question

How do we get the things we need?

PARKER'S PLAN

by Feana Tu'akoi

illustrations by Angel Mosquito

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Chapter 1

NEW INLINE SKATES

"Please, Dad," begged Parker. "Please may I have some inline skates?"

Parker's dad shook his head as he chopped the carrots. "We've been over this already," he said in a tired voice. "I keep buying you stuff that you use for a few weeks and then you get tired of it."

"But I will use them," said Parker, as he took the dinner plates out of the cabinet and headed for the table. "Miguel has some, and he can do great tricks on them. He's promised to teach me. Plus, we're planning to skate to school together every day."

His dad sighed and put the carrots into the pan with the other vegetables. "You say that now," he said, "but soon you'll be interested in something else. What about that tennis racket I bought you? What about the baseball mitt and the soccer ball? You never use any of those now." He put some noodles on to boil and gave the vegetables a stir.

"Anyway, I bought you some sneakers with wheels in them," his dad continued. "They still look brand new, so why don't you use those?"

Parker rolled his eyes as he set out the knives and forks. "They're for little kids. You can't do tricks on them," he complained.

"That's not what you said a few months ago," his dad reminded him with a grin.

Parker shrugged and went to fill up the water jug. "That was ages ago," he said. "I was way younger then, and I didn't know anything about skating."

"But it's not just the skates," said his dad. He added some meat to the pan and stirred to mix it in. "It's the safety gear. You'll need a helmet and a full set of pads. It's way too expensive. I can't afford it."



Parker thought for a moment about what his dad had said. "What if we wait until there's a sale?" Parker said. "Then we can get them for less money."

His dad shook his head. He drained the noodles and carried them to the table with the vegetables. "You know that money's been scarce since we bought a new car," he said, "so we need to be careful about what we buy."

Parker waited until they'd finished eating their dinner, and then he tried again. "What if I promise to use them every single day? Will you buy them then?"

"No," said his dad at last. "You're always changing your mind about what you're interested in, and I don't want to take a risk and waste any more money."



STOP AND CHECK

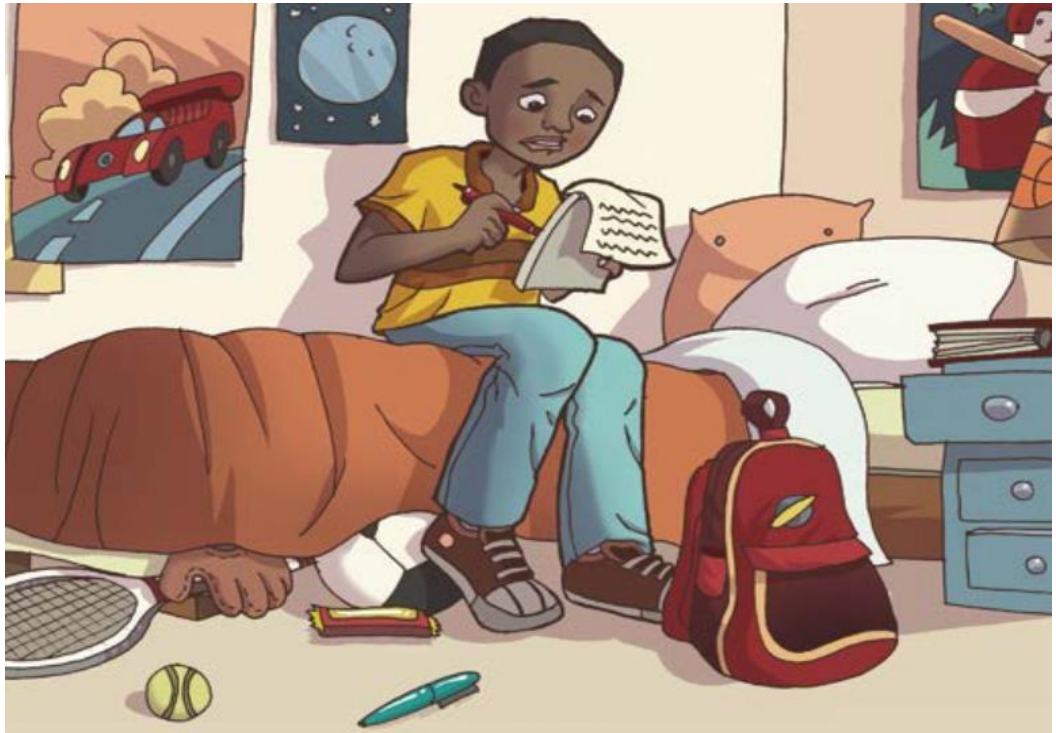
Why isn't Parker's dad willing to buy him inline skates?

Chapter 2
PLANNING PAYS

As soon as they finished doing the dishes, Parker stomped off to his bedroom. He wanted his dad to know that he wasn't happy about this decision.

He dumped his backpack onto the floor and threw himself across the bed. It just wasn't fair. Lots of kids had inline skates, and skating looked like so much fun. Why should he miss out?

But then he thought about what his dad had said. Parker sat up straight and grinned. His dad didn't actually say that he couldn't have the skates. All he said was that he wouldn't pay for them, so what if Parker could figure out a way to pay for them? He got out a pen and paper and began to write down some ideas.



Parker thought hard about how he could get the money. There were the 20 dollars that his grandma and grandpa would give him for his birthday, but that was months away. Aunt Amy always gave him some money when she came to visit, but she only visited a couple of times a year. Parker needed a solution now.

Maybe he could sell some of the things he didn't use anymore. But then he remembered that his baseball mitt was torn and his tennis racket had a broken string. No one would want to buy damaged gear.

"It's hopeless," he thought.

Parker suddenly had an idea. He could get a job! Why hadn't he thought of that before? If he earned extra money, he'd be able to buy the skates in no time. Parker was off the bed and out the door in a flash.

"Dad!" he cried as he skidded into the kitchen. "Wait till you hear my plan! I'm going to earn the money I need to buy those skates."

Parker's dad looked up from his book and smiled. "Great idea, Parker," he said. "If you work hard to earn the money to buy the skates, I'm sure you'll use them." He went to the closet and took out the mop. "You can start right away by working for me. How about mopping the kitchen floor?" he asked with a grin. "That's a dollar right there."



Before long, Parker had lots of jobs. It felt as if he spent all his spare time doing work for his dad. He washed windows, folded laundry, did the vacuuming, and he even cleaned the bathroom. Every time his dad paid him, the money went straight into his money box.

"This is great," he told his dad. "At the rate I'm going, I'll be able to buy my skates before the end of summer, and if I get really good, I might be able to join the roller hockey team!"

His dad laughed. "You're going to need more money, then," he said. "Roller hockey gear doesn't come cheap. You'll need lots of gear, so I'd better find you some more things to do!"

STOP AND CHECK

How does Parker's dad help Parker with his plan?



Chapter 3
THE SAVINGS ACCOUNT

A few days later, Talisha's mom came to school to talk to the class about her job as a bank manager.

"People trust us to take care of their money," she told the students. "We keep it safe and help it grow."

Parker raised his hand in an instant. He was desperate to make the money in his money box grow. "How do you make it grow?" he asked.

"If you put your money into a savings account," Talisha's mom explained, "the bank will pay you interest, so then you'll have more money than you started out with. And the more money you have in your account, the more interest you'll earn."

Parker's eyes widened. The bank would pay him for taking care of his money? That was unbelievable! He couldn't wait to tell his dad.

The next day after school, Parker's dad took him to the bank, where a bank employee named Enrique helped them. He asked Parker and his dad some questions and typed the information into a computer. Then he printed some forms for Parker and his dad to sign.

"Now," said Enrique, "are you going to make your first deposit?"



Parker handed him all the money from his money box.

"Do you know how much money you have there?" asked Enrique.

"Forty-one dollars and twenty-five cents," replied Parker proudly.

Enrique gave Parker some information about his account. Then he set him up for online banking and showed him how to check the account online.

His dad watched carefully. "I'll be able to transfer money directly into your account, Parker," his dad said.

As soon as they got home, Parker's dad went online to make a transfer into Parker's account. "I've paid what I owe you for this week straight into your account," he told Parker. "I'll do that every week from now on instead of giving you cash."

Parker grinned. He couldn't wait to check his account—it was growing already!

From then on, whenever he did a job, Parker asked his dad to transfer the money online. Soon he had enough to buy a cheap pair of inline skates. But his dad wouldn't let him buy them yet.

"Wait until you have enough money for the safety gear as well," he said. "That's really important."

STOP AND CHECK

How will having a bank account help Parker?



Chapter 4
WORKING FOR THE NEIGHBORS

A few weeks later, Parker was weeding the garden when his neighbor Mrs. Goldstein walked past.

"I didn't know you were interested in gardening, Parker," she said.

Parker straightened up and grinned. "I'm not," he said. "I'm interested in earning money so I can buy some inline skates."

Mrs. Goldstein looked over at her own garden. "Maybe you should work for me, too," she said. "I thought I'd have more time for gardening once I retired, but there's so much to do that I seem to have even less time now."

Parker beamed at her. "That would be great," he said. "But I'll have to check with my dad first."

Parker went to ask his dad. "That's fine," his dad said. "You can do jobs for other people, but only if we know them really well."

Parker decided to make fliers to give to their neighbors. The next day after school, he delivered his fliers. He was amazed. Within a day, he had all kinds of job offers.

For the next couple of weeks, he worked hard. He dusted apartments, walked dogs, and carried groceries. He read stories to toddlers, organized kitchen cabinets, and cleaned soccer shoes.



Most people paid him cash when he finished a job. Parker's savings grew quickly. Before long, he had enough to buy a great pair of inline skates, plus all the safety gear he needed.

His dad was impressed. "You've worked so hard—you must really want to learn inline skating," he said. "So, I've decided to pay for your roller hockey gear. If you're still interested," he added.

Parker stared at the floor, and his face turned as red as a beet. "Thanks, Dad," he mumbled. "But I don't think I'm going to buy skates after all."



"What?" said his dad. His eyebrows flew up his forehead. "But you and Miguel were going to skate to school together."

Parker grinned. "My money's growing so fast that if I keep saving, I'll be able to buy a BMX bike!" he said. "And I can always use my wheeled sneakers to get to school."

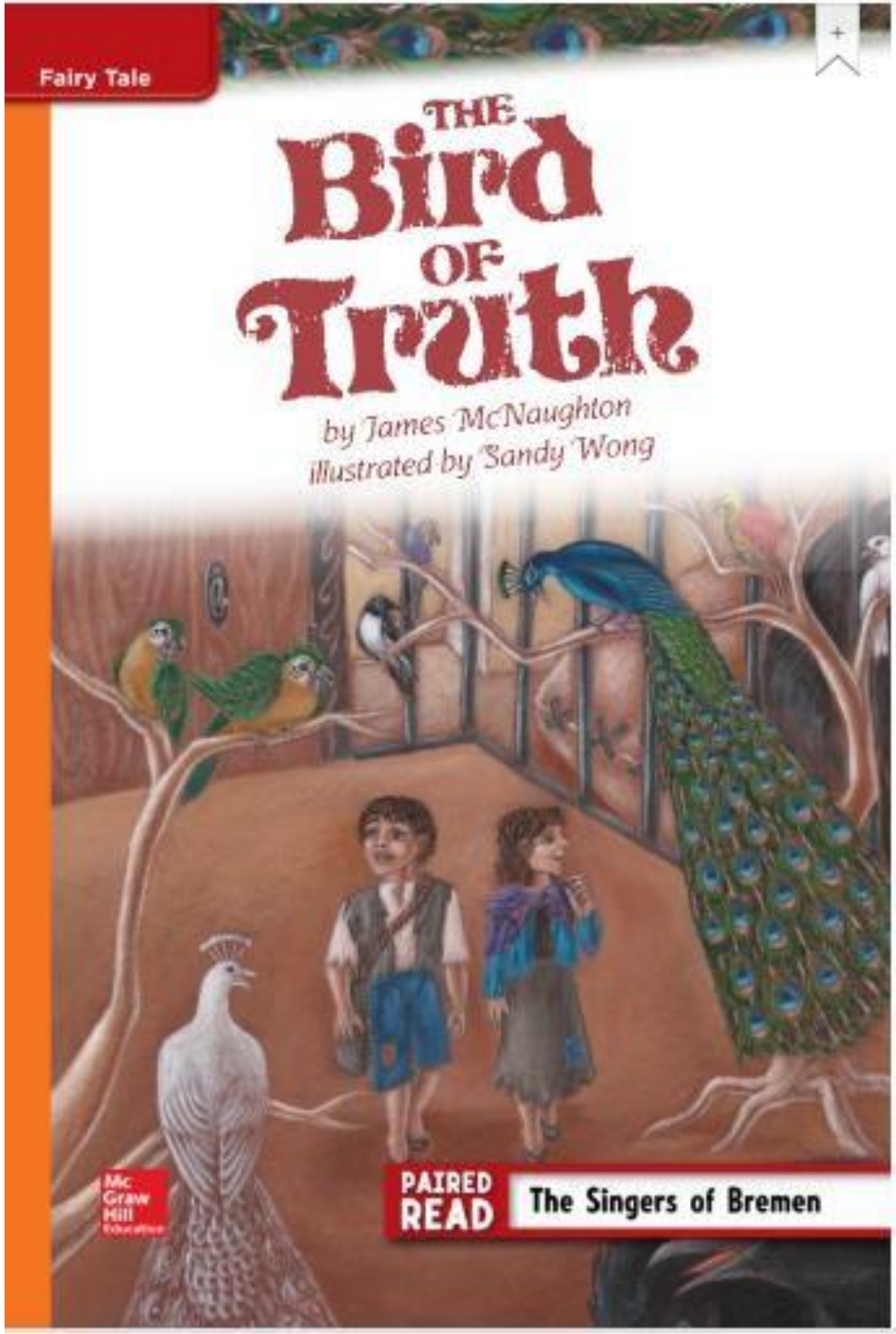
His dad's eyes grew wide. "I don't believe it," he said.

"Inline skates are cool," explained Parker, "but BMX bikes are better. My friend Ji Soo has one, and she can do all kinds of amazing things on it. It'll be expensive because I'll need to get all the safety gear, including body armor, but it'll be worth it. If I get really good, I might even be able to enter the state championships!"

"The state championships?" said his dad. He laughed. "Go for it!" he told Parker. "At this rate, you'll probably have the money before winter break!"

STOP AND CHECK

What does Mrs. Goldstein do to help Parker?



STRATEGIES & SKILLS

Comprehension

Strategy: Make Predictions

Skill: Compare and Contrast

Vocabulary

circumstances, consideration, consults, destiny, expectations, presence, reveal, unsure

Word Count: 2,423

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CHAPTER 1
TWINS

One day, a poor fisherman was casting his net into the river when he saw something in the water. It was shining in the sun like a diamond.

He lifted it out of the water. It was a cradle made of crystal. In the cradle were twin babies, a girl and a boy. They were carefully wrapped in soft blankets. The babies looked up at him and smiled.



Filled with pity, the man took the babies and cradle home to his wife. She threw up her hands in despair. "We already have eight children!" she cried.

"We can't put them back and let them drift out to sea," replied her husband.

The woman's heart softened at the sight of them. "You're right," she sighed, "and ten can eat as cheaply as eight."

The orphaned twins grew up to be good and gentle. Their adoptive parents came to love them even more than their own children, who were jealous and argumentative.

The couple's other children picked on the twins. To escape their teasing, the two spent hours alone on the banks of the river. The twins often brought bread they'd saved from their breakfast to feed the birds. In return, the birds taught the two children to speak their language.

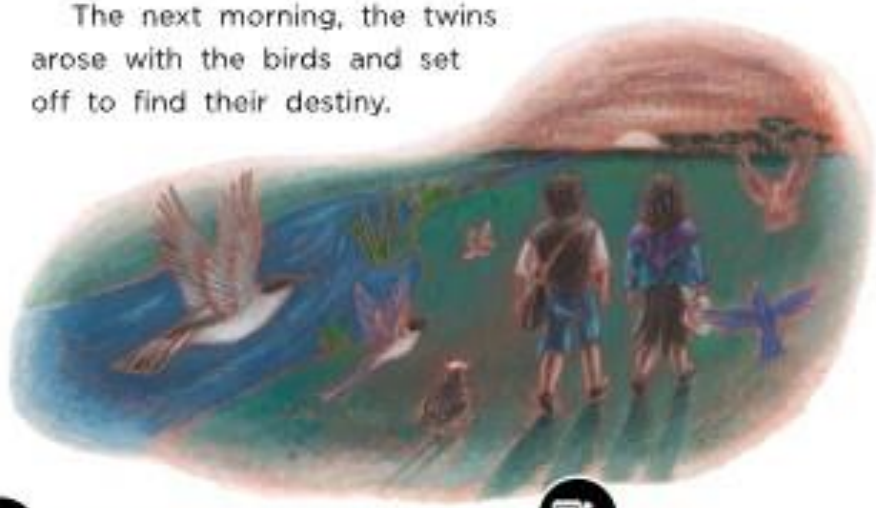


At first, learning a new way of speaking was very difficult, but the twins practiced whenever they could. Eventually they learned how to speak the language of birds.

The twins were kind and generous to their brothers and sisters. However, the other children continued to tease and taunt them. The eldest boy told them, "You think you're better than us, but at least we have a mother and father. You've only got the river, like the toads and frogs."

The poor twins didn't know what to say. The boy's insult made them very unhappy with their circumstances. They realized that their siblings would never accept them as part of the family, no matter how hard they tried. "We can't stay here any longer," they whispered to each other.

The next morning, the twins arose with the birds and set off to find their destiny.



STOP AND CHECK

How did the twins become part of the fisherman's family, and what made them decide to leave home?

CHAPTER 2
THE SWALLOWS

All day long, the twins walked along the river. They didn't see anyone else. By evening, the children were tired and hungry, but when they saw a small house ahead, their spirits rose. They imagined a warm and friendly welcome, a hot meal, and a comfortable bed. But their expectations were soon dashed. The house was dark and deserted.

The boy fought back hot tears. Trying to sound cheerful, he said, "At least we'll be able to lie down and sleep inside."

His sister smiled bravely. "Yes," she said after careful consideration. "We'll have a rest and decide what to do next."

They went inside and lay down in a corner to sleep. After they had been lying quietly for a while, some swallows flew in. The birds perched on the beams in the ceiling overhead. Unaware that the children knew their language, they began to chat freely. The swallows were a comforting presence, so the twins kept their eyes closed and listened.

"It sure is good to be back," said a swallow with a sigh.



"It's good to have you back," called the others happily.

"I tell you, life at the palace isn't all it's cracked up to be," said the first, who was sleeker than the other swallows.

"But wasn't it exciting seeing the king and hearing all the gossip?" asked another.

"Gossip?" said the first. "Birds don't like gossip!"

The birds all laughed, and the twins smiled at each other.

"So why did you leave?" asked one.

"It's peaceful here. There are too many cats in the palace! Enough about me, though—what's the news here?"

"Nothing ever happens here," said another. "Tell us the palace news!"

"Yes, yes," cried the others. "Tell us!"

When the country swallows had quieted down, the palace swallow began.

"You probably know that years ago, the king fell in love with a tailor's daughter. His nobles wanted him to choose a queen from one of their own daughters, so they tried to prevent the marriage. The king didn't listen to their protests. He went ahead with the wedding anyway. A few months later, the king went away to war. When he returned, the nobles told him that the queen had given birth to twins, but they had both died. They told the king that his wife was sick with grief and had shut herself in a tower in the mountains."

"Was that the truth? Were the twins dead?" asked the country swallows.

"Of course not!" replied the palace swallow. "The babies were alive. The nobles didn't want any heirs to the throne, so they imprisoned the queen and told the gardener to kill the babies. The gardener couldn't do it. Instead, he put them in a crystal cradle, then he laid it in the river. The babies were found by a fisherman, who raised them as his own."

STOP AND CHECK

What do the children learn from the swallows?

CHAPTER 3
THE BIRD OF TRUTH

When the twins heard about the crystal cradle, they gasped in astonishment. Their adoptive parents had told them the story of how they were found. "I can't believe it!" whispered the girl to her brother.

"If we hadn't learned the language of the birds, we might never have found out," he replied.

One of the country swallows said, "Why don't the children return to their father and set their mother free?"

"They won't be able to prove who they are," said the palace swallow. "The only way things can ever be put right is if the king consults the Bird of Truth. But she's trapped in a castle. A giant keeps her locked in a large cage with the Birds of Bad Faith. They make so much noise that she cannot be heard."

"Where is this castle?" cried the swallows.

"I don't know," replied the palace swallow sadly.

"Oh dear, oh dear, oh dear," said the birds.

The twins knew what they had to do.



At daybreak, the twins set off to find the giant's castle. They wanted to free the Bird of Truth. The countryside became bleaker and lonelier, but there was still no sign of the castle. They threw themselves to the ground in despair. Suddenly, they heard movement in the tree above them. It was a turtledove.

"Oh dove," the boy cried, using the bird's language, "please help us. We need to find the castle where the Bird of Truth is caged."

"Follow the wind," said the dove. "Today it's blowing toward the castle. If it pities you, it will blow steadily."

They thanked the bird and followed the wind. It blew and blew, guiding the children until they came to a large and gloomy castle. The doors were wide open, but the children still hesitated. The castle was as quiet as a grave. They were very frightened. Unsure of what to do next, they hid behind a lone olive tree near the castle door.



"What shall we do?" the boy asked the girl.

"What shall you do?" repeated an owl that was sitting above them in the tree.

The twins leaped for joy and said softly in the bird's language, "We've come to free the Bird of Truth. How should we do it?"

"There is an aviary in the great courtyard," said the owl. "It is filled with many brightly colored birds. Each one will call to you, saying it is the Bird of Truth. Be sure to ignore them. They are the Birds of Bad Faith. Take only the small white bird hidden in a corner. Now go while the giant is asleep, for he sleeps only 15 minutes a day!"

The twins ran quickly through the open doors and into the great courtyard. There they saw an enormous aviary. As soon as they stepped inside, the birds began to make a deafening noise. Peacocks, parrots, ravens, and all sorts of other birds called out. The birds all claimed to be the Bird of Truth.

The twins ignored the uproar. They walked among the birds until they found a small white bird in a corner.



The Bird of Truth was hemmed in by crows. The girl reached down and picked up the bird.

"Thank you, children. I have been waiting for someone to set me free," she said.

The girl gently placed the Bird of Truth inside her jacket. Then the twins ran as fast as they could out of the aviary and through the castle gates, followed by the screams of the Birds of Bad Faith.



Terrified that the giant would be awakened by the noise, the children kept running until the castle was out of sight.

News that the Bird of Truth was free spread throughout the kingdom. This made the king's wicked nobles and courtiers nervous. They were afraid she would reveal the truth about the queen and their children to the king. So they set traps to catch her and trained eagles and falcons to hunt her.

CHAPTER 4
TOGETHER AGAIN

The king spent most of his time alone in the palace. He didn't get much news. Usually he only heard what the nobles wanted him to hear. But this rumor was so strong that even the king knew about it. He proclaimed that whoever found the Bird of Truth should bring her to him immediately.

The twins heard about the king's proclamation and headed straight for the palace. However, at the entrance, the courtiers stopped them.

All of a sudden, the Bird of Truth flew out of the girl's jacket and into the king's chamber through an open window. She bowed respectfully. "Your Majesty, I am the Bird of Truth," she said. "The boy and girl who brought me here are being held at the gate by your courtiers."

The king ordered the children to be brought to him at once, and he asked them for an explanation.

"Please, Your Majesty," said the girl, "the Bird of Truth will explain *everything*."

And so the Bird of Truth told the king about the nobles' wicked plot and how they had imprisoned the queen and tried to kill his children.

When the king learned that he'd been misled, he took the twins, and together they hurried to the tower where the queen was imprisoned.

The queen was as pale as marble, but when she saw her husband and children, the color returned to her face, and she became as beautiful as ever.

"It's so good to see you," she said to the king. "Why did you wait so long?"

The king looked shocked. "Every month I received a letter from you telling me to stay away."

"The nobles lied to you," she said. "I wrote every day, asking you to come."

The king wiped away tears as he embraced the queen and their children. "No one will ever separate us again."



When the royal family returned to the city, they were greeted by cheering crowds. A great banquet was held in celebration. The fisherman and his wife sat in a position of honor next to the grateful king and queen. The wicked nobles fled the kingdom and never returned. And so it was that the family was reunited and lived happily ever after.

STOP AND CHECK

How are the twins reunited with their parents?

Respond to Reading

Summarize

Use important details from *The Bird of Truth* to summarize how the twins learned important information. Your graphic organizer may help you.

Event →	Outcomes
→	
→	
→	
→	

Text Evidence

1. What features of the story tell you that *The Bird of Truth* is a fairy tale? **GENRE**

2. How are the events in Chapter 2 similar to and different from the events in Chapter 3? **COMPARE AND CONTRAST**

3. What does the simile "quiet as a grave" on page 9 mean? **SIMILE**

4. Write about the events that happened after the twins were born and the events that helped them return to their parents. Use details from the text to help you compare and contrast the events. **WRITE ABOUT READING**

STRATEGIES & SKILLS

Comprehension

Strategy: Summarize

Skill: Author's Point of View

Content Standards

Social Studies

Civics and Government

Vocabulary

anticipation, defy, entitled, neutral, outspoken, reserved, sought, unequal

Word Count: 1,358**

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**The total word count is based on words in the running text and headings only. Numerals and words in captions, labels, diagrams, charts, and sidebars are not included.

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Genre Biography

Essential Question

What can people do to bring about a positive change?

Jane Addams

A Woman of Action

by Jane Buxton



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Chapter 1

Early Years

Do you know what you want to do with your life? Jane Addams thought about this a lot when she was a child.

Jane was born in 1860, and she grew up in Cedarville, Illinois. Her mother died when Jane was young. Jane's father took care of her and her sisters. Mr. Addams was a rich man, and he was also a **philanthropist**, a person who helps others.



Even as a young girl, Jane wanted to make a difference in the world.

Jane Addams Collection, Swarthmore College Peace Collection

Jane learned from her father to be kind and to accept others' differences. She loved to learn and to read.

Jane saw that some people were rich and others were poor. She could see that the rich and the poor had unequal lives. She wanted all people to be treated fairly.

Jane wanted to be a doctor. Her father wanted her to get married and have a family. Jane did not want to get married, but she also did not want to defy her father.



Jane's father, John Addams, was a successful landowner, miller, and banker.

Jane Addams Collection, Swarthmore College Peace Collection

Addams had a good education. She studied at Rockford Female Seminary. She was good at reading, writing, and public speaking.



Addams (back row) made some good friends at school.

When Addams was 21 years old, her father died. Addams was devastated. She decided to study to be a doctor, but she was too sad to study. Then she became very ill, so she gave up college and her dream of becoming a doctor.

Addams traveled overseas after she left college. She went to London and saw many people living in **poverty**. She wanted to help, but what could she do?

She visited a settlement house in London called Toynbee Hall. It was called a settlement house because the young men who worked there settled there to live. The young men were educated, and they wanted to help the people in the neighborhood. They offered classes in singing, reading, and drawing.

Addams was excited. She wanted to set up a settlement house, too!

STOP AND CHECK

Who taught Jane Addams to help other people?

Extreme Poverty

Addams visited a London market. It was the end of the week, and the market was closing. The rotten food was being sold. The food was bad, but Addams saw thin people dressed in rags trying to buy it. These people were *very* poor—they were living in extreme poverty. Jane never forgot what she saw.

Chapter 2

Addams Takes Action

Addams returned to Illinois, where she and her friend, Ellen Starr, looked for a house to rent. The two friends wanted to set up a house like Toynbee Hall. They would use the house to improve the lives of the poor.

They found an enormous, run-down old house in a poor neighborhood in Chicago. The house was called Hull House.

Addams and Starr knew that some Americans wanted to learn about the differences between the rich and poor. They wanted to teach these Americans more about the lives of poor people. Classes at Hull House could help teach people about equality.



Hull House was originally owned by a wealthy businessman, Charles Hull.

Bettmann/Getty Images

Hull House would offer poor local people education and classes in music. This kind of opportunity was usually reserved only for the wealthy.

Addams and Starr wanted educated women and men to live at Hull House. They wanted these people to get to know their neighbors, who were neither rich nor educated.

Addams had money of her own, which she used to fix up Hull House. She also sought money from other wealthy citizens to help keep the house running.

Life in the City

Addams and Starr set up Hull House at a time when many people were coming to America from Europe. These people were looking for a better way of life.

People moved to Chicago to find work in factories. Women, men, and even children worked long days and nights. They worked in places that were not safe or healthy. They were not paid enough money. The bosses (the business owners) became very rich.

With great anticipation, Addams and Starr opened Hull House in 1889. Many people wanted to help her, and they moved into Hull House, too.

The local people liked to meet at Hull House. They could take many different kinds of classes, including English classes. Hull House had a theater, a library, an art gallery, a kitchen, a gymnasium, and a music school. People could also go to Hull House if they needed help with child care, or if they needed a doctor or a lawyer.

People liked what Addams was doing, and they set up settlement houses in other cities.



Addams started clubs for children.

Valuing Traditions

Addams set up a labor museum at Hull House. At the museum, older immigrants used spinning frames and weaving looms. It was a way for them to share with others their knowledge of traditional arts and crafts. Addams wanted people to realize that the skills of these older immigrants were still valued.

Addams was kind and goodhearted. She was an outspoken **advocate** for the poor, and she talked to many people about the importance of helping others. Addams became famous in the United States.

Hull House needed money to keep going. Addams wrote a book about her life, called *Twenty Years at Hull House*. She was a good writer. The money she made from her books was used to support her work.



Addams wrote a total of 13 books.

STOP AND CHECK

What kinds of things could people do at Hull House?

Chapter 3

Changing the World

During Addams's life, there were many poor women, men, and children working in factories. The work was unsafe. They worked long hours and were not paid enough. However, if they did not work, they would starve.

Three children were hurt at a factory. One of them died. The machine they used was dangerous. Addams talked to the factory owner. She asked him to put a protective cage over the machine. She was shocked when the owner refused.



Children often worked in factories like this one that made fabric.

Children Are Cheap

Most factory bosses did not want to change child labor laws. They thought children should understand the value of hard work. They said that widows needed the money their children could earn. Child labor also meant it was cheaper to run their businesses. Child labor was finally banned in 1938.

Addams could not help poor people by herself. She realized that poverty would not go away unless the laws were changed.

Addams wanted to make things better for workers. She believed women should work only eight hours each day. She wanted a better justice system for young people. She worked hard to make school free for all children.

She talked to politicians about changing the laws. Many people agreed with Addams. They worked together to make changes. It took a long time, but the laws were **reformed**.

Addams knew that to make positive changes in the world, she needed to get people to change their attitudes. Addams was known for her work helping the poor. She also helped people who were treated unfairly because of their race. Addams helped set up the American Civil Liberties Union (ACLU) and the National Association for the Advancement of Colored People (NAACP). Both organizations still exist today.

Everyone Is Responsible

Addams believed that everyone has a responsibility to help others. Addams spent her life working to make positive change. She worked to help women, children, and ethnic minorities. She also worked for better health care, for the poor, and for peace. She asked, what is more important than having "faith in new possibilities and the courage to advocate them?"

Addams also tried to change people's attitudes about war. She thought that war was wrong. She even tried to stop World War I.

Addams talked to many people about the importance of peace. Many people supported her. They set up women's peace organizations with Addams. They created the Women's Peace Party, the International Congress of Women, and the Women's International League for Peace and Freedom.



The Women's Peace Party delegates sailed on this ship to the International Congress of Women in The Hague in 1915.

Not everyone agreed with Addams. Some people wrote her nasty letters. Some newspapers criticized her work. Addams didn't stop, though. She knew she was entitled to say what she believed. She kept working for change and for peace.

Jane Addams is one of America's best-loved and most well-respected leaders. She spent her whole life working to make positive changes for people.

Addams was courageous. She was often sick, but she never stopped working for what she believed. She never gave up her goal of helping others. In 1931, Addams received the Nobel Peace Prize in recognition of her life's work.

STOP AND CHECK

Which groups of people did Addams try to help?



Addams was the first American woman to receive the Nobel Peace Prize.

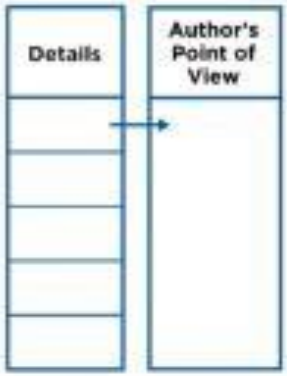
Time Line: Jane Addams

1860	born, Cedarville, IL
1881	graduates from Rockford Female Seminary
1887	visits Toynbee Hall
1889	founds Hull House
1909	helps found NAACP
1910	publishes *Twenty Years at Hull House*
1915	helps found the Women's Peace Party and organizes the International Congress of Women
1919	helps found the Women's International League for Peace and Freedom
1920	helps found the American Civil Liberties Union
1931	receives the Nobel Peace Prize
1935	dies, Chicago, IL

Respond to Reading

Summarize

Use important details from *Jane Addams: A Woman of Action* to summarize what you learned about Addams. Your graphic organizer may help you.

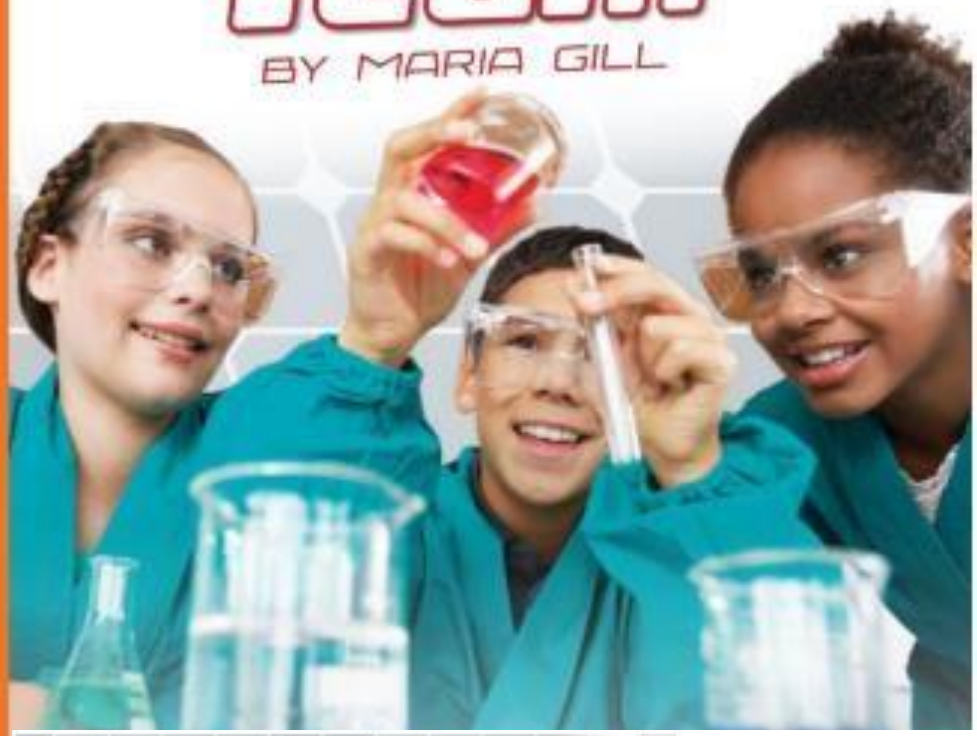


Text Evidence

1. What type of text is *Jane Addams: A Woman of Action*? How do you know this? **GENRE**
2. What does the author think about Addams? How can you tell? **AUTHOR'S POINT OF VIEW**
3. The word *unequal* on page 3 includes the prefix *un-*. The prefix *un-* means "not." Use the meaning of *un-* and context clues to help you define *unequal*. **PREFIXES AND SUFFIXES**
4. Write about the author's views on Addams's work to ban child labor. Look for details in the text that support your answer. **WRITE ABOUT READING**

The Power of a Team

BY MARIA GILL



Introduction

Have you noticed that it's easier to figure out really hard problems when you work with other people? Sometimes they see things that you can't see. They look at the problem in a new way.

People with different skills produce great results when they collaborate. Physicists, biologists, and geologists can research different parts of the same problem. By sharing their ideas, they can reach

Genre Expository Text

Essential Question

What benefits come from people working as a group?

The Power of a Team

BY MARIA GILL

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Introduction

Have you noticed that it's easier to figure out really hard problems when you work with other people? Sometimes they see things that you can't see. They look at the problem in a new way. Other times, you have done all you can on your own. You need someone else to finish things off. Science is just the same.

Scientists around the world work in teams to answer difficult questions. Some teams might look for a cure for a disease. Others might try to discover more about space.



Scientists often work together.

People with different skills produce great results when they collaborate. Physicists, biologists, and geologists can research different parts of the same problem. By sharing their ideas, they can reach solutions a lot faster.

A TEAM EFFORT

Australian scientist Howard Florey was one of the first to gather a team of scientists together. He was trying to find better ways of treating infections, but this problem was too big to solve alone.

The team wanted to learn more about penicillin. Each scientist researched the uses of penicillin in his area. By sharing their discoveries, the team members found a cure for many common illnesses.

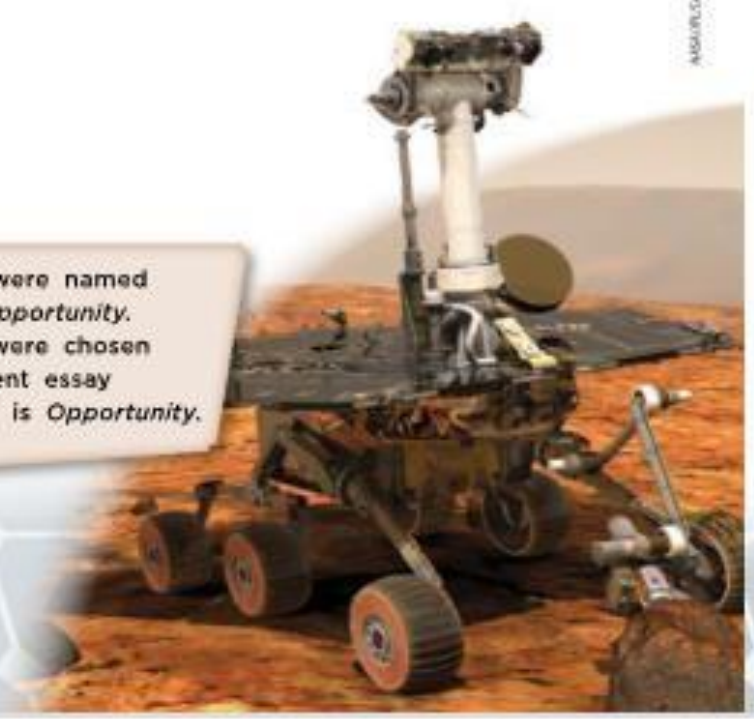


Howard Florey and his team saved thousands of lives.

Chapter 1
New Frontiers

Scientists at **NASA** work in teams. One team worked on two robots to explore Mars. They spent three years planning, building, and launching them. The robots were made to rove, or travel, across the planet. They could scan rocks and soil and send the data back to Earth.

Before the robots left Earth, the team had to find a place to land them safely. Experts studied images of the planet. They picked two sites that looked like they used to have water. The robots lifted off in 2003.



The robots were named *Spirit* and *Opportunity*. The names were chosen from a student essay contest. This is *Opportunity*.

The robots took about seven months to reach Mars. When the rover called *Spirit* landed, the engineers noticed a problem. An **air bag** from *Spirit*'s **lander** was blocking the rover's path. The driver in the control room knew what to do. He turned *Spirit* sideways and drove it down a side ramp instead.

The second rover, *Opportunity*, also landed safely. Then the team could relax. More than three years of hard work had paid off.

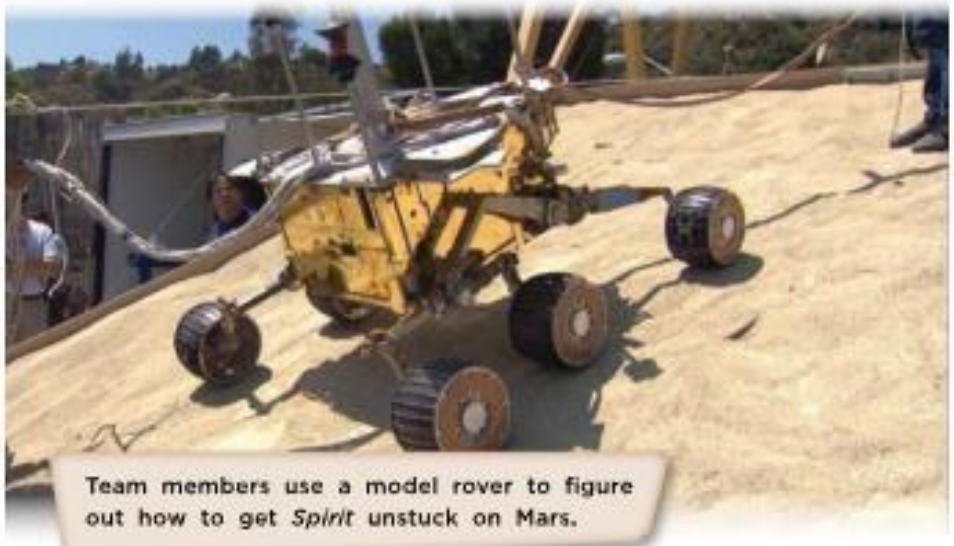


The Mars Exploration Rover team gets the news that *Spirit* has arrived safely.

During the next few months, *Spirit* traveled over rough terrain. Its front wheel burst, but the team found a way to keep the rover going. They sometimes drove *Spirit* in reverse.

Then *Spirit* got stuck. The team made a sand trap on Earth to mimic *Spirit*'s problem. Experts studied a model of the stuck rover. Then the team tried out possible solutions with the model.

The attempts to free *Spirit* led to an exciting discovery. As *Spirit*'s front wheel dug into the soil, it uncovered a bright substance called silica. Because silica was there, scientists knew that there had once been water on Mars.



Team members use a model rover to figure out how to get *Spirit* unstuck on Mars.

Opportunity spent its first two years roving around Mars. Its next job was to explore a large **crater**. The data *Opportunity* has sent back will help geologists to discover more about the history of Mars.



Opportunity sent back images of craters on the surface of Mars.

Spirit stopped sending signals back to Earth in 2010. *Opportunity* is still roving across Mars.

A NASA EXPERIENCE

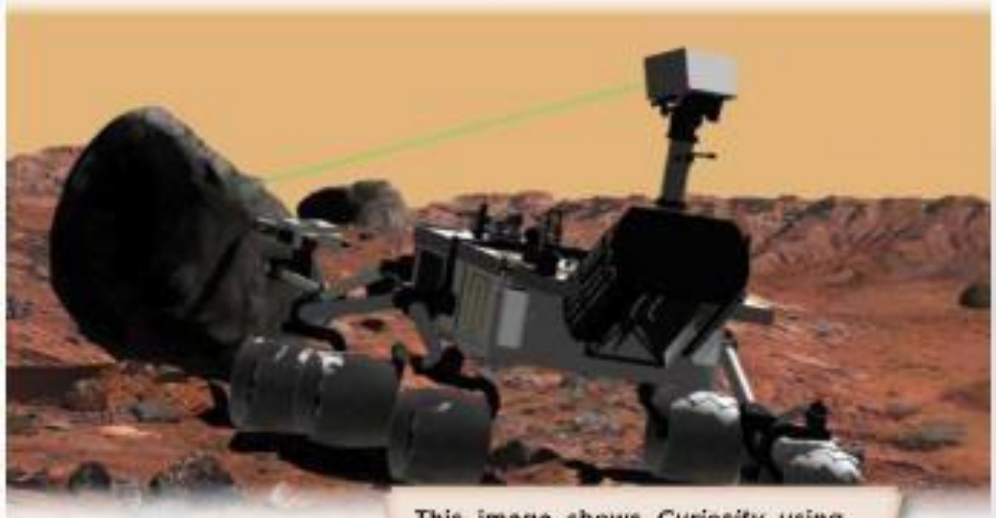
Thirteen teams of high school students and teachers worked with the Mars Exploration Rover scientists during the rovers' launches and landings. The students woke up at 4:00 A.M. for team meetings. They had jobs to do and attended press conferences. The students saw how everyone on the team worked together.

The next rover mission sent a new type of robot to Mars. The Mars Science Laboratory is the size of a car.

The first model of this new robot is named *Curiosity*. It left Earth on November 26, 2011. It landed on Mars on August 6, 2012. It has tools for collecting and analyzing rock and soil samples. It has found evidence that water once flowed on the surface of Mars.

STOP AND CHECK

What did scientists do when *Spirit* got stuck?

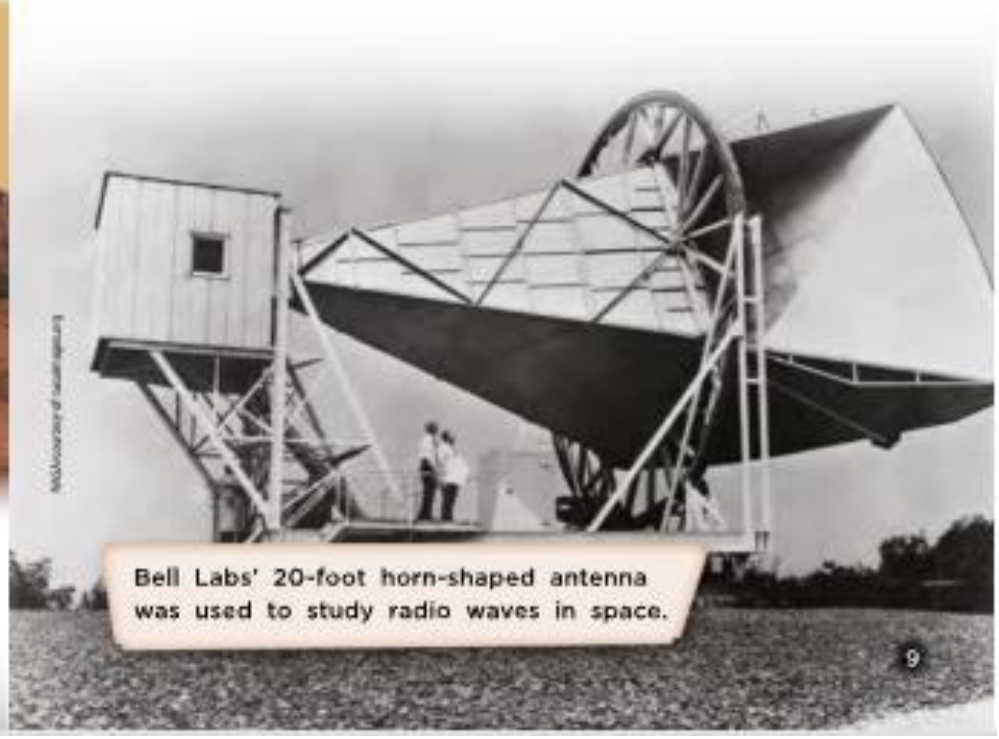


This image shows *Curiosity* using a laser to vaporize rocks. Then the dust can be analyzed by special instruments.

Chapter 2
New Discoveries

In 1964, two scientists at Bell Labs discovered something they weren't looking for.

Arno Penzias and Robert Wilson were using a huge antenna to study radio waves in space. They kept hearing an annoying sound. It was like static. Pointing the antenna in another direction didn't make the noise go away. Then the scientists noticed bird droppings on the antenna. They shooed away the birds and cleaned the antenna, but that annoying hiss didn't go away.



Bell Labs' 20-foot horn-shaped antenna was used to study radio waves in space.

Scientists at Princeton University were studying the **big bang theory**. This theory says the universe was formed in a huge explosion. Low-level **radiation** might have been left behind after the explosion. The scientists at Princeton and Bell Labs talked to each other. They realized the antenna must be picking up low-level radiation.

That hissing sound is now called cosmic microwave background radiation. The Bell Labs scientists won the **Nobel Prize** for their accidental discovery.

A SMALL INVENTION WITH A BIG IMPACT



In the 1940s, telephones didn't work very well. A group of scientists began trying to make telephone signals stronger. For two years they had no success. Then they wrapped gold foil around a small plastic triangle. The triangle was touching a mineral called germanium. The scientists sent a signal through the device, and the signal came out much stronger on the other side. The team had succeeded!

The team did some more work on their design. They built a transistor. Today, transistors are used in almost all electronic equipment.

The scientists at Bell Labs are still working hard. They are looking for ways to make it easier for people to communicate.

When you surf the Internet or make a phone call, you use a communication network. A network is equipment that is linked together. When you punch in a friend's phone number, the network picks up this signal. It carries the signal to your friend's phone.

That's similar to how nerves in your skin work. If you are cold, your nerves send a message through your nervous system telling your brain this.



Humans love to communicate.

Our nerves send lots of signals to our brains all the time. They don't need our help. They just do it. However, communication networks need our help. They can jam when they get too many signals. This can happen when too many people use their cell phones at the same time.

Bell Labs wants to invent flexible networks. These would handle lots of signals automatically, just like our bodies do. Around the world, team members who are experts in math, computing, and engineering are working on this idea.

STOP AND CHECK

What did the Bell Labs scientists accidentally discover?

People want to connect anytime, anywhere as fast as possible.



Chapter 3 New Beginnings

Sometimes people work together for personal reasons. That's the case with The Miami Project.

Marc Buoniconti was 19 when he was hurt in a football game. His spine was damaged. He was paralyzed from the neck down. Marc's father, Nick, wanted his son to walk again. Nick helped start The Miami Project. Marc is now president of The Miami Project.



Marc and Nick Buoniconti have raised millions of dollars for spinal research.

SPINAL CORD INJURIES

Signals from our nervous system travel through the body. They go up the spinal cord to the brain. The brain also sends signals down the cord to the body. The spinal cord is a long bundle of nerves protected by small bones called vertebrae. When we see someone make a great baseball catch, the brain sends a signal down our spinal cord telling our hands to clap. Marc's spinal cord was so badly damaged that his brain no longer communicated with his body.



Scientists around the world work on The Miami Project. Some are figuring out how to prevent spinal cord damage. Others are looking at how to repair the spinal cord. Another team is working on how to help patients recover.

The team thinks that cooling the body down right after a spinal cord injury helps prevent more damage. They tested this by having doctors lower their patients' body temperatures. Then patients were slowly warmed up again.

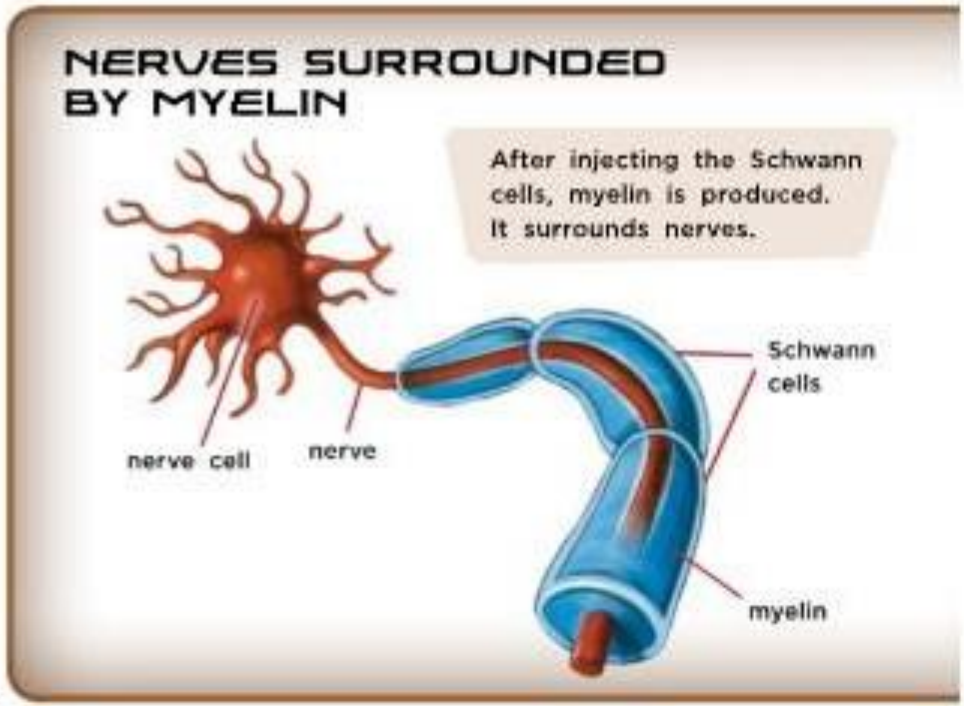
After one year, these people had more movement than is usual for people with spinal cord injuries. Scientists will do a much bigger **trial** with hundreds of patients.



Cooling the body very soon after a spinal cord injury reduces swelling and might help the patient recover sooner.

The Miami Project's scientists think that injecting special cells called Schwann cells into a damaged area of the spine might help fix it. These special cells have an important function. They make a substance called myelin. The myelin covers the nerves and helps the nervous system communicate.

Researchers tested their idea on paralyzed rats. The rats got back 70 percent of their ability to walk. Scientists are now planning a trial with human patients.



The Miami Project's scientists are also working on techniques to reduce patients' pain.

The scientists found that using more than one treatment can also help recovery. Teams in separate research areas are working toward one goal. They are dedicated to helping people with spinal injuries to walk again.

STOP AND CHECK

What discoveries did The Miami Project scientists make?



Dr. James McLurkin wants to get robots to cooperate and to respond to their environment.

ROBOT TEAMS

What if we sent a team of 200 robots to Mars, instead of just two? How much more could we learn? Dr. James McLurkin would like to find out. He works on multi-robot swarms. These are groups of robots working together as a team.

The robots use infrared transmitters to share information. This helps the group work together. Dr. McLurkin says that his work was inspired by the teamwork of honeybees. They share information and respond to their environment as a group.

Conclusion

When scientists and experts work together, they can get better results.

The noise that scientists at Bell Labs heard was a missing piece in the big bang puzzle. It took hundreds of people to put the rover robots on Mars.

By sharing ideas, scientists can make faster progress. One day, they may be able to bring rocks back from Mars. Or they may help paralyzed people to walk again.

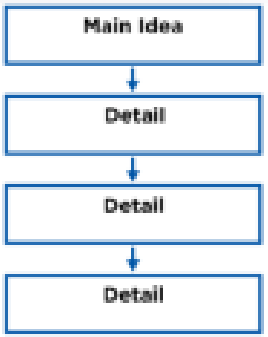


Kids can experience the power of a team, too!

Respond to Reading

Summarize

Use important details from *The Power of a Team* to summarize the selection. Information from your graphic organizer may help you.



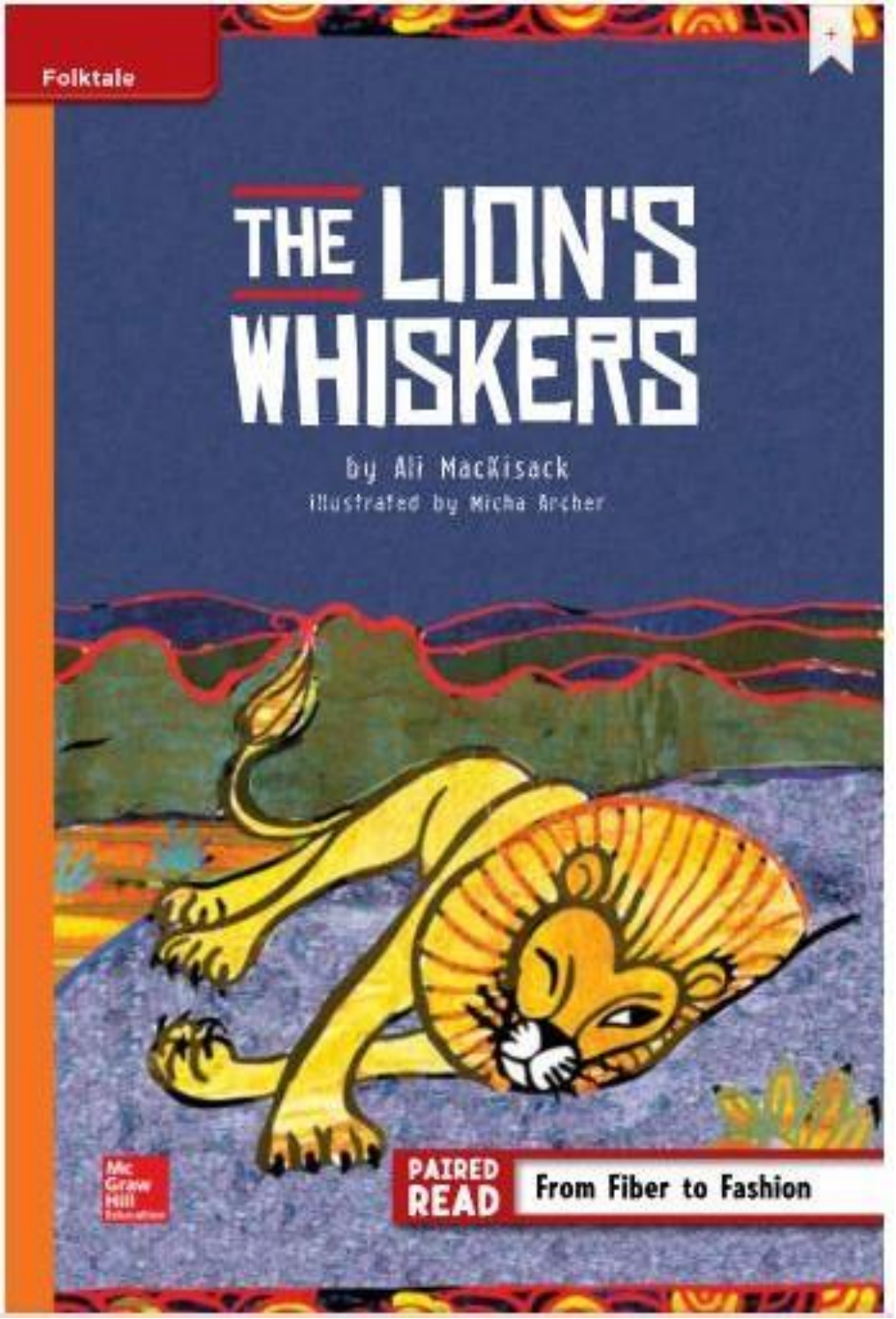
Text Evidence

1. How can you tell that *The Power of a Team* is expository text? **GENRE**

2. What is the main idea of the first paragraph on page 4? Give key details to support your answer. **MAIN IDEA AND KEY DETAILS**

3. Look at the word *terrain* on page 6. It includes the Latin root *terr*, which means "earth." Use the meaning of this word part and context clues to figure out what *terrain* means. **LATIN ROOTS**

4. Write the main idea of each paragraph on page 14. Give examples of key details that helped you determine each main idea. **WRITE ABOUT READING**



STRATEGIES & SKILLS

Comprehension

Strategy: Make Predictions

Skill: Theme

Content Standards

Science

Physical Science

Vocabulary

assuring, detected, emerging, gratitude, guidance, outcome, previous, pursuit

Word Count: 1,984**

**The total word count is based on words in the running text and headings only. Numerals and words in captions, labels, diagrams, charts, and sidebars are not included.

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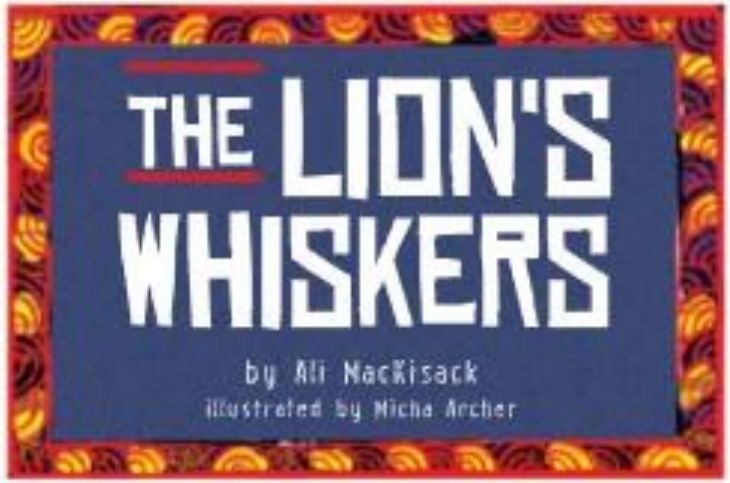
11 12 13 14 15 DOC 22 21 20 19 18

F

Genre Folktale

Essential Question

When has a plan helped you accomplish a task?



Chapter 1
A Rocky Start 2

Chapter 2
A Cautious Plan 6

Chapter 3
Words from the Wise 12

Respond to Reading 16

PAIRED READ **From Fiber to Fashion** 17

Focus on Genre 20

Chapter 1

A ROCKY START

There was once a woman named Alitash. She lived in the highlands of Ethiopia. Alitash was a kind, clever, and hardworking woman. She loved children and longed for a family of her own.

One day, Tesfa, who was a merchant in Alitash's hometown, asked her to marry him. Alitash said yes. She was very happy to marry Tesfa.

Tesfa had a son named Dawit from his first marriage, so Alitash gained a son as well as a husband. This pleased her, but Dawit was not happy at all. His mother was dead, and he missed her. He did not want Alitash to take his mother's place. Dawit cried himself to sleep at night and refused to be comforted by Alitash.

Alitash wanted to be a good mother to Dawit. She cooked his favorite meals. She made him new clothes from the finest cotton cloth. She brought him special treats and bought him new shoes. No matter what Alitash did, Dawit turned his back on her and cried for his own mother.

Alitash wanted Dawit to be happy. She also wanted him to love her. She understood his deep sadness and knew that only love could heal his heart.

Because Tesfa was a merchant, he often traveled with his caravan of mules. While he was gone, Alitash could see that Dawit missed his father as well as his own mother. In spite of his loneliness, Dawit did not turn to Alitash for comfort. She began to think that the boy would never grow to love her at all.



Sad and worried, Alitash went to see a wise woman who was a leader in the village.

She poured out her story to the wise woman and asked for her guidance. "Perhaps you could make a special potion that would make my stepson love me?" Alitash suggested.

The woman sat in silence for a long time. She stared into the fire, then lifted her head to meet Alitash's hopeful gaze.



"I can prepare a potion as you have asked," she said. "But it is a very difficult task. To make a love potion, you must first bring me something very special."

"Anything—I will bring you anything you need," said Alitash, assuring the woman of her determination. "I will bring you whatever you need because I love my stepson very much. I long for him to love me back."

"Very well, then," replied the wise woman. "You must go into the desert and bring back three whiskers from a lion. The lion must be alive, and you must do this on your own."

Alitash gasped. She knew how dangerous this was. She knew she could be killed. With a heavy heart, she left the wise woman's hut.

STOP AND CHECK

Why does Alitash go to see the wise woman?

Chapter 2

A CAUTIOUS PLAN

That night, Alitash couldn't sleep. She paced the floor, trying to decide what to do. She wanted Dawit to love her, but getting three whiskers from a lion seemed impossible. She tried to think of a plan, but her thoughts just chased each other around in circles.

Should she build a trap? Should she try to trick the lion in some way? Could she give him something to make him sleep?

She didn't know what to do. How could she get close enough to the lion to remove three whiskers? And the wise woman said that she must do it on her own!

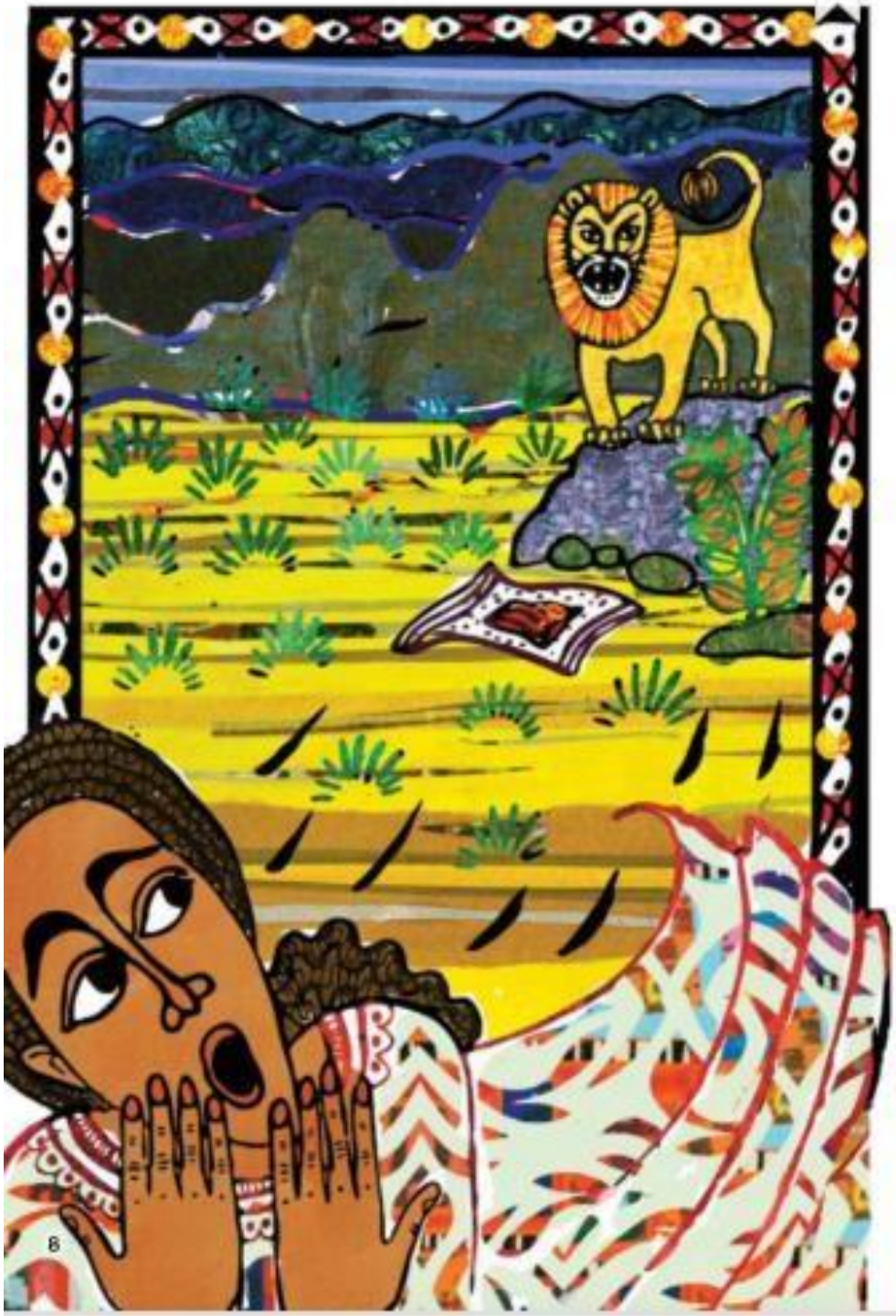
However, Alitash was both clever and determined. As the sun began to rise, a plan was emerging in her head. She decided to put the plan into action that very day.



Later that morning, Alitash wrapped a piece of meat in a cloth and carried it far into the desert. She headed for a place called Lion Canyon where she knew there were lions.

As she approached the place, she saw a lion sprawled on top of a rock. He was huge, and he was watching her. Her legs were unwilling, but her determination was strong. She continued walking forward. She did not take her eyes off the lion. And the lion did not take his eyes off her.





When she was still some distance away, the lion rose to his feet, threw back his head, and roared. Alitash dropped the meat and ran. The sound echoed around the rocky walls of the canyon and followed Alitash as she fled.

She had no idea if the lion was in pursuit, but she didn't look back. She didn't stop running until she was home again.

The next day, Alitash ventured into the desert again. She was carrying another piece of meat.

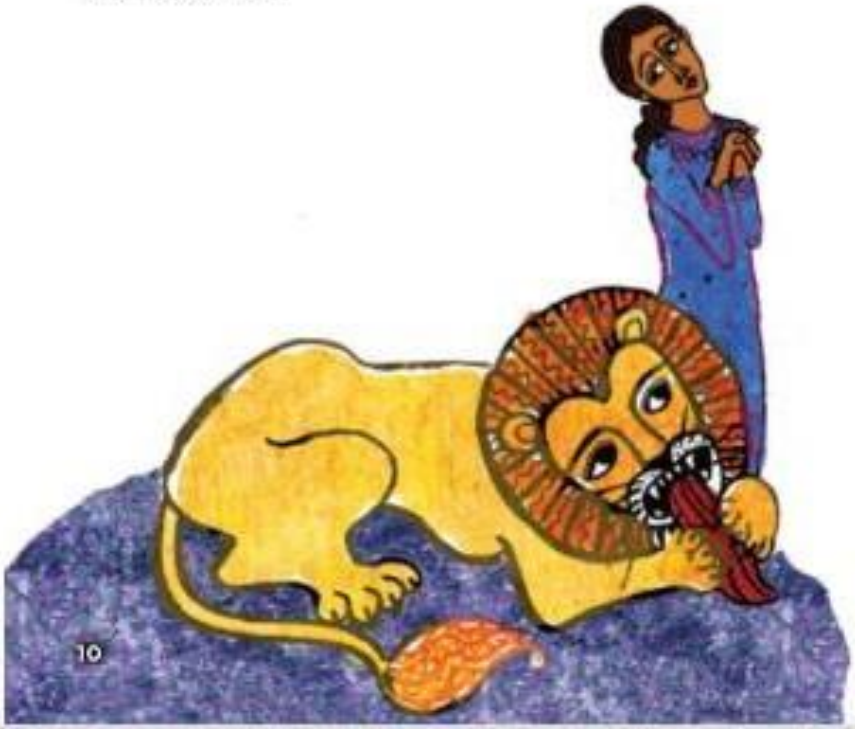
The lion was sprawled out on the same rock he had been on the previous day. This time, Alitash got a few steps closer before he stood up and roared.

She flung the meat onto the ground and fled. But this time, she glanced back over her shoulder as she left the canyon. The lion had jumped down from his rocky outcrop. He was beginning to eat the piece of meat.

Over the following months, Alitash walked into the desert every day. Every day, she took meat for the lion. And every day, she drew closer and closer to him before he roared.

Eventually she didn't run away when she dropped the meat. Instead, she retreated more slowly and watched the lion while he ate. Each time, she stood a little bit closer to where the lion ate. Soon the lion began to accept her presence.

At last the day came when the lion let her stand beside him while he was eating. Alitash knew it was time to see if her plan would bring the outcome that she hoped for.



The next day, Alitash hid a small, sharp knife in a pocket in her dress before she went to Lion Canyon. Her heart was pounding as the lion approached her. When he dropped his head to eat the meat, Alitash made her move.

Her knife sliced through the whiskers at the side of the lion's face. The lion jerked his head up, and the whiskers fell to the ground. He snarled at Alitash. His powerful jaws were only an arm's length away from her.

She leaped back, fearing that this was the end. But the lion snarled again, snatched up the meat, and carried it to his rock to finish eating.

Alitash bent down and picked up the whiskers from the dust.

"Thank you, mighty lion," she whispered. "Now Dawit will finally love me."

STOP AND CHECK

How does Alitash carry out her plan?

Chapter 3

WORDS FROM THE WISE

Alitash was full of excitement as she cooked the evening meal for her family that night. When she handed a bowl to Dawit, she smiled at him, and she thought she detected a tiny smile in return.

"What strong powers these lion whiskers have," she thought. "It seems they are working even before the potion has been made."



Early the next morning, Alitash went to see the wise woman. When she pulled the three lion's whiskers from her pocket, the woman looked amazed. She stared at the whiskers for a long time. Suddenly she snatched the whiskers and threw them in the fire.



"Why did you do that?" cried Alitash. "I did what you told me to do. I brought you the whiskers of a living lion, and I did it all on my own!"

The wise woman put a hand on Alitash's arm. "You are a brave and clever woman," she said. "You do not need a potion. By bringing me the lion's whiskers, you have shown me that you already have what you need to gain your stepson's love."

"But how can that be?" Alitash asked.

"You do not need a potion to win your stepson's love," said the wise woman. "You will win his love with what is inside you. Tell me, what was the first thing you did to try to get these whiskers?"

"First I thought of a plan to get the lion to let me near him," answered Alitash.

"And how did you approach the lion?" the wise woman asked.

"Slowly, with patience and respect," was Alitash's reply.

"Now tell me why you think your plan succeeded," the wise woman said.

"Because I followed my plan step by step, and I stayed focused on my goal."

"And what was it that drove you?" asked the wise woman.

"It was my love for Dawit," replied Alitash. "And my wish for him to love me back."

"You see?" said the wise woman. "You already know in your heart what you must do to win the boy's love. Approach him the same way as you did the lion—with patience and respect. Take things slowly, go step by step, and stay focused on your goal. In time, just like the lion, the boy will learn to trust you. Love will come if you are patient."

So Alitash left the wise woman's hut without a potion, but with gratitude and a new understanding.

She kept the wise woman's words in mind, and little by little, her relationship with Dawit grew warmer and closer.

One day Dawit asked Alitash to make him some clothes for an important festival. She smiled to herself and thought of the day when she had finally stood within reach of the lion's whiskers, and she knew that one day Dawit would love her as much as she already loved him.



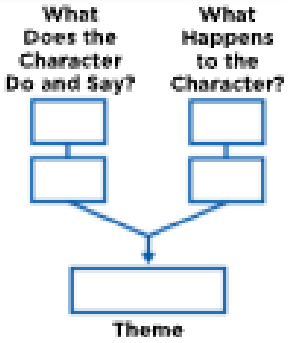
STOP AND CHECK

What happened when Alitash took the lion's whiskers to the wise woman?

Respond to Reading

Summarize

Use important details from *The Lion's Whiskers* to summarize how Alitash's plan helped her accomplish her goal. Your graphic organizer may help you.



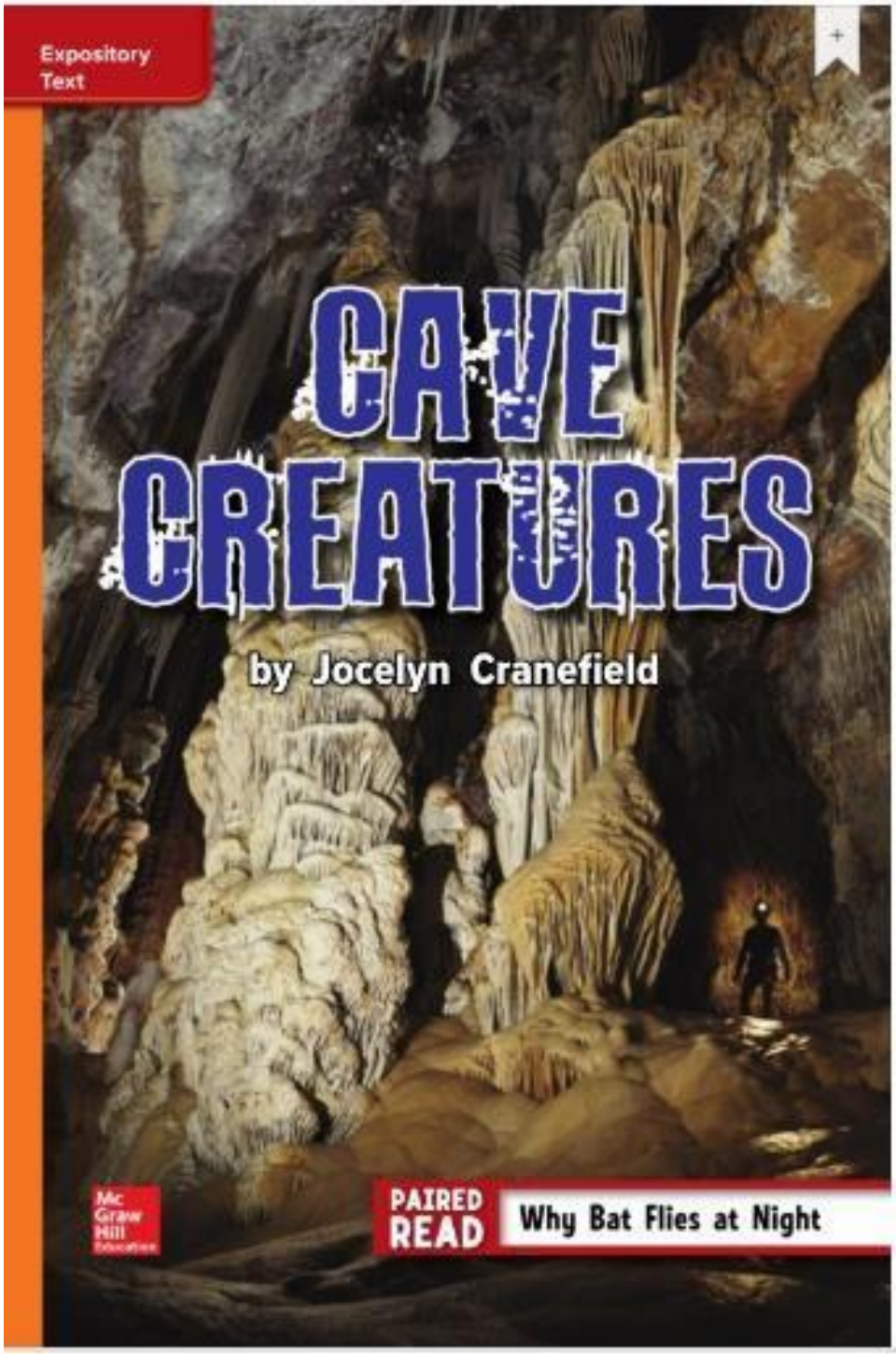
Text Evidence

1. How do you know that *The Lion's Whiskers* is a folktale? Give details from the story. **GENRE**

2. What is the theme of *The Lion's Whiskers*? How does Alitash's success in getting the lion's whiskers support the theme? **THEME**

3. Explain why "her thoughts just chased each other around in circles" on page 6 is an example of personification. **PERSONIFICATION**

4. Reread the wise woman's words on page 14. Write about the message the author communicates through the wise woman's words. **WRITE ABOUT READING**



STRATEGIES & SKILLS

Comprehension

Strategy: Ask and Answer Questions

Skill: Cause and Effect

Vocabulary

adaptations, agile, caches, dormant, forage, frigid, hibernate, insulate

Content Standards

Science

Life Science

Word Count: 1,434**

Photography Credit: MICHAEL NICHOLS/National Geographic Creative

**The total word count is based on words in the running text and headings only. Numerals and words in captions, labels, diagrams, charts, and sidebars are not included.

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Genre Expository Text

Essential Question

How are living things adapted to their environment?

CAVE CREATURES

by Jocelyn Cranefield

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INTRODUCTION

Caves are found all over the world. They often go a long way in from the entrance and go deep below Earth's surface. At first a cave might seem to be just a dark, empty space.

If you shine your flashlight inside the cave, you probably won't see anything. But if you go inside, you probably won't be alone.



Caves can be on the coast, in a forest, or under a desert. They can be made out of limestone, marble, or lava. Some caves have beautiful stone formations, and others are full of smelly, poisonous gases. However, every cave in the world provides a home for animals.

Lechuguilla Cave in New Mexico is famous for its beautiful mineral formations.

Some animals only visit caves and don't live there. Other animals live in caves all the time. Many of these creatures have adaptations to help them live in a cave. Adaptations are special features that help a living thing survive in a particular place.

CAVES OF ALL SIZES

More than 50,000 natural caves have been found in the world. Some caves are single tunnels, but others are huge systems of caves that connect with each other. The deepest cave we know of is the Krubera Cave, near the Black Sea. It is at least 7,188 feet deep. Compare its depth with the heights of some tall buildings, below. The longest cave is Mammoth Cave in Kentucky. It runs for more than 350 miles underground.

The Krubera Cave



CHAPTER ONE

FROM THE ENTRANCE TO THE TWILIGHT ZONE



As the sun sets, these bats come out of their cave to hunt.

Some animals visit caves to shelter or to sleep. They stay near the cave entrance. These animals are called trogloxenes, or cave visitors. Bats, skunks, raccoons, and snakes are trogloxenes. They spend part of their lives in caves, but they need to go outside to get their food.

The Mexican free-tailed bat is a trogloxene. It is **nocturnal**. During the day, it roosts in **colonies** that hang from cave roofs. At dusk it flies out of the cave to catch insects.

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Most bats have an adaptation to help them find their way in a dark cave. This adaptation is called echolocation. The bat makes high-pitched squeaking sounds and listens for an echo. The echo helps it find out where things are.

In some places, bats hibernate in caves in the winter. They don't stay asleep all winter. They usually wake up from their dormant state for a short time every 15 to 30 days.

In cold climates, bears may also spend the winter in caves. The air is warmer inside the cave. It helps to insulate animals from the cold.



The Virginia big-eared bat hibernates in caverns in West Virginia.

Bears go into a deep sleep in winter. Some scientists say this is a kind of hibernation.





A glowworm can turn off its light if it hears a noise.

Deeper inside the cave is the twilight zone. It is cool and damp here. Many of the creatures living here are troglophiles, or cave lovers. They may live in caves for all of their lives, but they can also survive outside. Some spiders, earthworms, beetles, frogs, and crickets are troglophiles.

The twilight zone has little food or light. The animals living here have adaptations so they can survive in these conditions. They use senses other than sight, such as hearing or touch, to get food and find their way around.

The glowworm is also a troglophile. It is the larva of an insect called a fungus gnat. Its adaptation is lighting up its abdomen to attract prey. The glowworm spins a sticky web and then switches on its light. It caches tiny insects in its web until it needs to eat a meal.

The New Zealand cave weta is a kind of cricket that lives on the undersides of cave roofs. It has **antennae** that are seven times longer than its body. It uses its antennae to help it move around and find food. This agile insect also has long, strong legs. It can jump more than 6 feet!



The New Zealand cave weta's long legs help it to get away from danger.

ADAPTING TO SILENCE

In places where there is little food, it's useful to be quiet to avoid being eaten! The African cave cricket does not chirp to communicate. Instead, it uses its wings to send out doughnut-shaped puffs of air called vortices. Only other cave crickets can feel these vortices.



What do animals eat in a cave? Plants can't grow in the dark, but food comes into caves in different ways. Rain and streams wash in twigs, leaves, seeds, and insects. There is plant material in the droppings of animals such as bats and cave crickets. These are good sources of food for cave creatures.

After molds, **fungi**, and **bacteria** break down all this material, microscopic animals can eat it. These microscopic animals are eaten by larger cave dwellers. Then the larger animals are eaten by other predators.

CAVE LIONS



Scientists learned about these big cats from their very old bones.

Some scientists believe the European cave lion lived between 12,000 and 40,000 years ago. It was much bigger than today's lions.

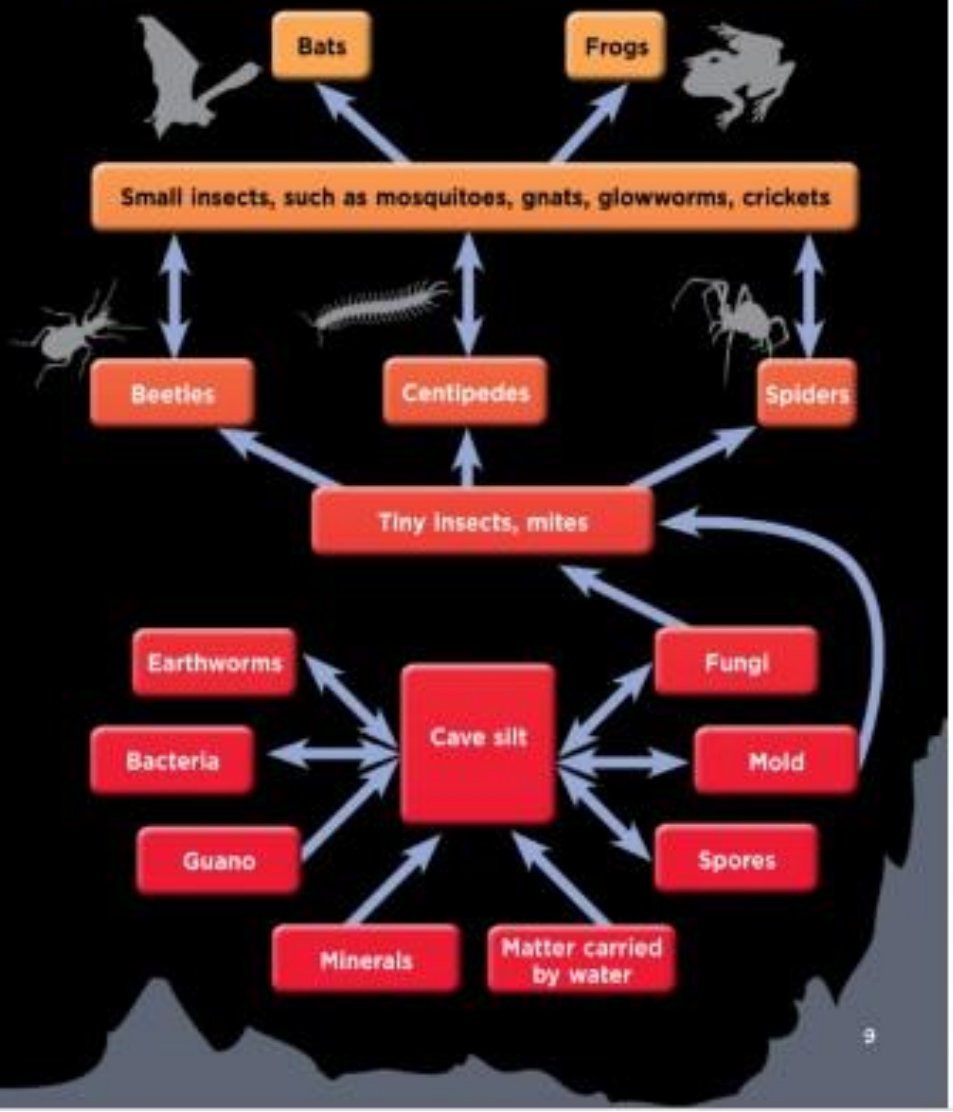
Skeletons of animals that die in caves can be preserved for a long time. This is because conditions inside caves rarely change.

STOP AND CHECK

Why do animals need adaptations to live in caves?

CAVE FOOD WEB

The nutrients in guano, or bat droppings, and other materials are broken down and recycled by some cave creatures and microscopic organisms. These creatures are eaten by larger predators.



CHAPTER TWO

DARK AND SURPRISING PLACES



This pseudoscorpion has long pincers but no eyes.

Imagine going deep underground into the dark zone of a cave. There is no light and no plants grow. It is dark and silent, and the air is still.

Life in the Dark Zone

Animals that live in the dark zone all the time are called troglobites, or cave dwellers. They have adaptations, such as small bodies, long limbs, and long antennae. They can pick up small vibrations or smells. These adaptations help them to move about and forage for food in the pitch-black space.



This insect, called a dipluran, has long antennae and two tails.

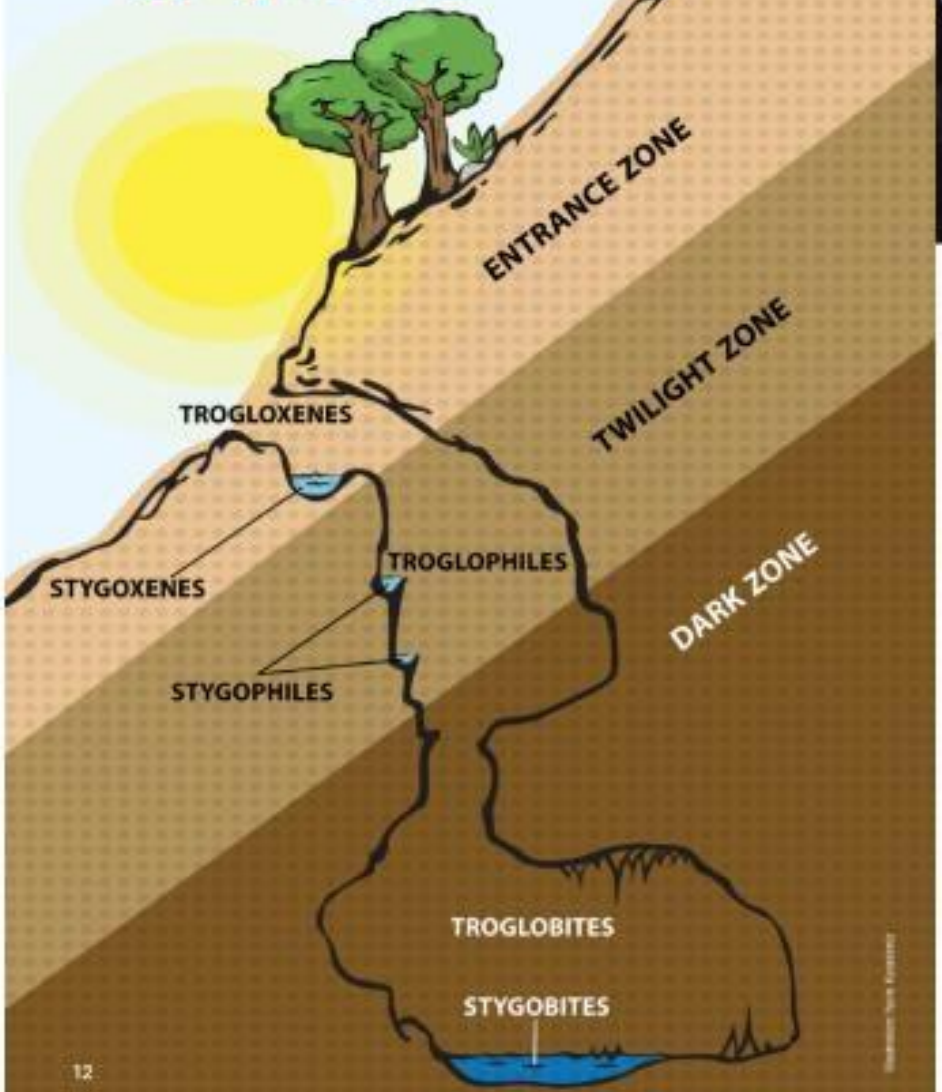
Many troglobites have different adaptations from the animals that live above ground. Animals living in the dark zone usually have tiny eyes, or no eyes at all, because eyes are not useful if you live in the dark.

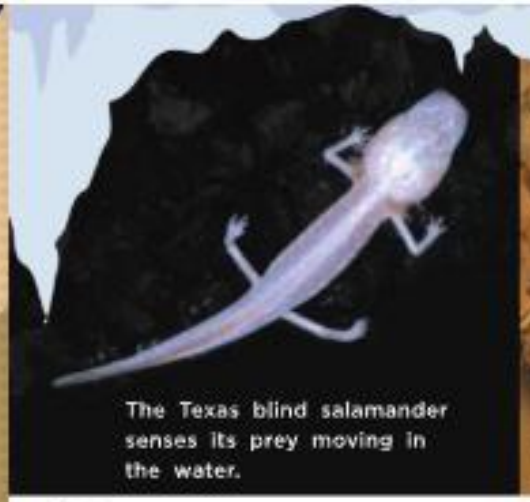
Most animals living above ground have coloring in their skin called pigment. Pigment is an adaptation to protect animals from the sun or give them camouflage. Pigment is not needed in the dark zone of a cave. As a result, some animals in caves have skin that is nearly see-through. These cave dwellers could not live for long in the world outside.



This adapted daddy longlegs has long legs and a tiny body. It is also blind.

THE ZONES OF A CAVE





The Texas blind salamander senses its prey moving in the water.



This cave crayfish has no pigment.

Aquatic Cave Dwellers

Many caves have networks of underground streams and ponds. There may be **aquatic** animals living in and around the water.

Aquatic cave dwellers that can't live outside the cave are called stygobites. Stygobites include blind, pale-colored fish, **salamanders** with no eyes, and **crustaceans** with no pigment.

Stygoxenes are aquatic creatures that come from outside the cave. Stygophiles are water lovers that may spend time outside the cave as well as inside it.

(©Daria Ferreira/Dorling Kindersley/Getty Images, ©Tom Whitaker/Alamy Stock Photo)

Extreme Cave Dwellers

Cave creatures have been found in unusual places. In 2001, tiny stygobites were found in caves under the desert in Western Australia. They looked like shrimp.

Some caves provide even more hostile conditions. These caves are too cold, too hot, or too toxic for most cave creatures to live in. Yet microscopic forms of life have been found living in these places. Animals that are adapted to live in extreme conditions are called extremophiles.



Extremophiles live in Greenland's frozen ice caves.

Greenland's ice caves have frigid conditions. Most animals could not survive in them. But microscopic creatures have been found there. These creatures can cope with the caves' sub-zero temperatures.

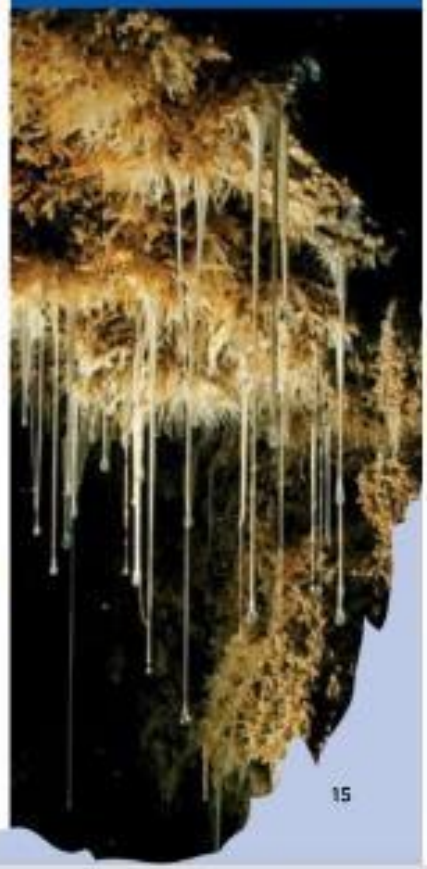
Cueva de Villa Luz in Mexico is another extreme cave. It has fumes of a toxic gas called hydrogen sulfide. Scientists call the microbes that live on this gas names such as snottites, blue goo, and slime balls!

STOP AND CHECK

How are creatures living deep inside caves different from animals living above ground?



The water bear can live in many difficult environments.



This type of bacteria in Mexico's Cueva de Villa Luz lives on toxic gas.





CONCLUSION

Many cave creatures couldn't survive in our world. The adaptations that help them live underground would make it dangerous for them above the surface.

How do we know which features of cave creatures are adaptations? And how do we learn how these adaptations help animals to survive? Scientists try to find the answers to these questions. They go underground to observe and photograph cave animals to find out about them.

Animals living deep inside caves can't travel easily to other places. Scientists have found some rare species. They may bring the creatures up to Earth's surface. The scientists mimic the conditions of caves in laboratories and examine the creatures closely.

An amazing range of creatures has been found in caves. But we don't understand everything about these animals and the way they live. There are still many things to learn about life in caves.

Respond to Reading

Summarize

Use the most important details from *Cave Creatures* to summarize the selection. Your graphic organizer may help you.

Cause → Effect
→
→
→
→

Text Evidence

1. What text features help you identify *Cave Creatures* as an expository text? **GENRE**

2. Reread page 11. Why do some creatures in caves have tiny eyes or no pigment? What words in the text help you find causes and effects? **CAUSE AND EFFECT**

3. What does the word *hostile* mean on page 14? How do context clues in the paragraph help you figure it out? **PARAGRAPH CLUES**

4. Write about two cave creatures and the adaptations they have. How does each adaptation help the creature live in the cave? Use details from the text in your writing. **WRITE ABOUT READING**